AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 6 · STUDENT LESSON

Forward passes by hand

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Compute in order. Find the first error. Retest the repair.

Audit complete forward traces using independent expectations and revealing regression cases.

Keep one habit: Five-row trace → scalar first-error picture → four submitted-trace clinic → zero/order probes → independent different network.

MAIN LESSON · SECTION 1

Where is the first value that stops following the rule?

> Debug in dependency order rather than patching only the final answer.

Learning goals

I can record scores and activations in **dependency** order.

I can locate the first incorrect value in a forward **trace**.

I can separate a final **output** from **evidence** of model accuracy.

Recall the earlier information path

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- In Week 5, what does the **output** layer receive?

> **My starting explanation**

End goal

By the end, I can explain forward passes by hand using a **trace**d example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

This week you will practise:

Record scores and activations in **dependency** order
</details>

Locate the first incorrect value in a forward **trace**

Separate a final **output** from **evidence** of model accuracy

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Fix the network and vary the input

> A trace names preactivation,activation,output contributions and bias separately.

Complete network and dependency order

Use a new one-**input** network with two hidden units. Its scores are z1 = 2x - 1 and z2 = -x + 2. Apply ReLU to obtain h1 and h2. The **output** is y = h1 - 2h2 + 1. Parameters stay fixed while the **input** changes.

The **input** x is an abstract number; these are arithmetic examples, not measurements of children.

A forward pass evaluates this network in **dependency** order. Compute both scores from x, then both activations, then the **output** contributions h1 and -2h2, and finally add bias 1.

A **trace** table makes an omitted bias or a sign error easier to locate than a final answer alone.

Three complete starting traces

x	z1 and z2	h1 and h2	y
0	-1 and 2	0 and 2	-3
1	1 and 1	1 and 1	0

x	z1 and z2	h1 and h2	y
3	5 and -1	5 and 0	6

Zero output and the optional class rule

At x = 1 the **output** is exactly zero. If we later declare class 1 for y >= 0, equality belongs to class 1. Applying ReLU in the hidden layer does not make the final **output** nonnegative.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Check every stage against an independent calculation

> Repeated output is repeatable behavior;expected intermediate values check the declared computation.

Complete worked input and five-row trace

For x = 0, the first score is -1, which ReLU changes to 0. The second score is 2, which ReLU leaves at 2. Therefore y = 0 - 4 + 1 = -3. Negative **output** is allowed because this **output** layer uses identity.

Main x · z1=2x−1 · z2=−x+2 · h1=ReLU(z1) · h2=ReLU(z2) · Output contributions h1 /−2h2 /bias · y

> **Main x:** 0
>
> **z1=2x−1:** -1
>
> **z2=−x+2:** 2
>
> **h1=ReLU(z1):** 0
>
> **h2=ReLU(z2):** 2
>
> **Output contributions h1 /−2h2 /bias:** 0 /−4 /+1
>
> **y:** -3

> **Main x:** 0.5
>
> **z1=2x−1:** 0
>
> **z2=−x+2:** 1.5
>
> **h1=ReLU(z1):** 0
>
> **h2=ReLU(z2):** 1.5
>
> **Output contributions h1 /−2h2 /bias:** 0 /−3 /+1
>
> **y:** -2

> **Main x:** 1
>
> **z1=2x−1:** 1
>
> **z2=−x+2:** 1
>
> **h1=ReLU(z1):** 1
>
> **h2=ReLU(z2):** 1
>
> **Output contributions h1 /−2h2 /bias:** 1 /−2 /+1
>
> **y:** 0

> **Main x:** 2
>
> **z1=2x−1:** 3
>
> **z2=−x+2:** 0

> **h1=ReLU(z1):** 3
>
> **h2=ReLU(z2):** 0
>
> **Output contributions h1 /−2h2 /bias:** 3 /0 /+1
>
> **y:** 4

> **Main x:** 3
>
> **z1=2x−1:** 5
>
> **z2=−x+2:** -1
>
> **h1=ReLU(z1):** 5
>
> **h2=ReLU(z2):** 0
>
> **Output contributions h1 /−2h2 /bias:** 5 /0 /+1
>
> **y:** 6

Runnable fixed-parameter forward trace

```
def trace(x):
    z = [2 * x - 1, -x + 2]
    h = [max(0, value) for value in z]
    y = h[0] - 2 * h[1] + 1
    return z, h, y

for x in [0, 1, 3]:
    print(x, trace(x))
```

This optional Python uses only numeric **input**s. Compare every returned component against the table.

The same code being run twice is a repeatability check; comparing it with a separately calculated expectation is a stronger check of correctness.

Connect a visual step to its actual value

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Keep the model fixed while testing a new trace

> Locate the first disagreement,then recompute everything that depends on it.

Current bias-omission case

- At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured submitted-trace error

- Change: A submitted x = 0 **trace** uses h=-1 and y = 5.

▼ Check after predicting

Prediction: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Separate the first error from its later consequences

> Use the named model’s formula before comparing two traces.

Complete pictured model and scope

Audit a forward pass by writing every value at its correct stage.

Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

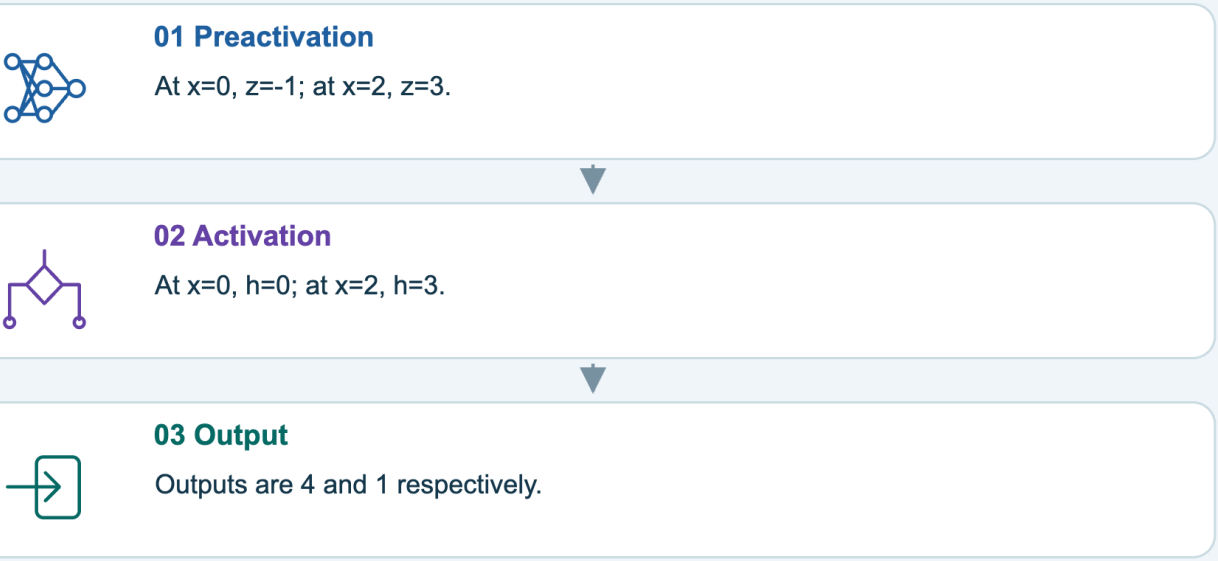
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate scalar-hidden model:z=2x−1, h=ReLU(z), y=−h+4. At x0, z−1, h0, y4; at x2, z3, h3, y1. The submitted x0 trace h−1, y5 first fails at activation because ReLU(−1)=0. Its final arithmetic follows the wrong h; repair h and then y. The main model has two hidden units and output y=h1−2h2+1, so its x0 output is−3, not 4. Keep the formula beside each trace.

Trace the scalar stages

AI ACADEMY / AI-5/6 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Preactivation	At x=0, z=-1; at x=2, z=3.
2. Activation	At x=0, h=0; at x=2, h=3.
3. Output	Outputs are 4 and 1 respectively.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

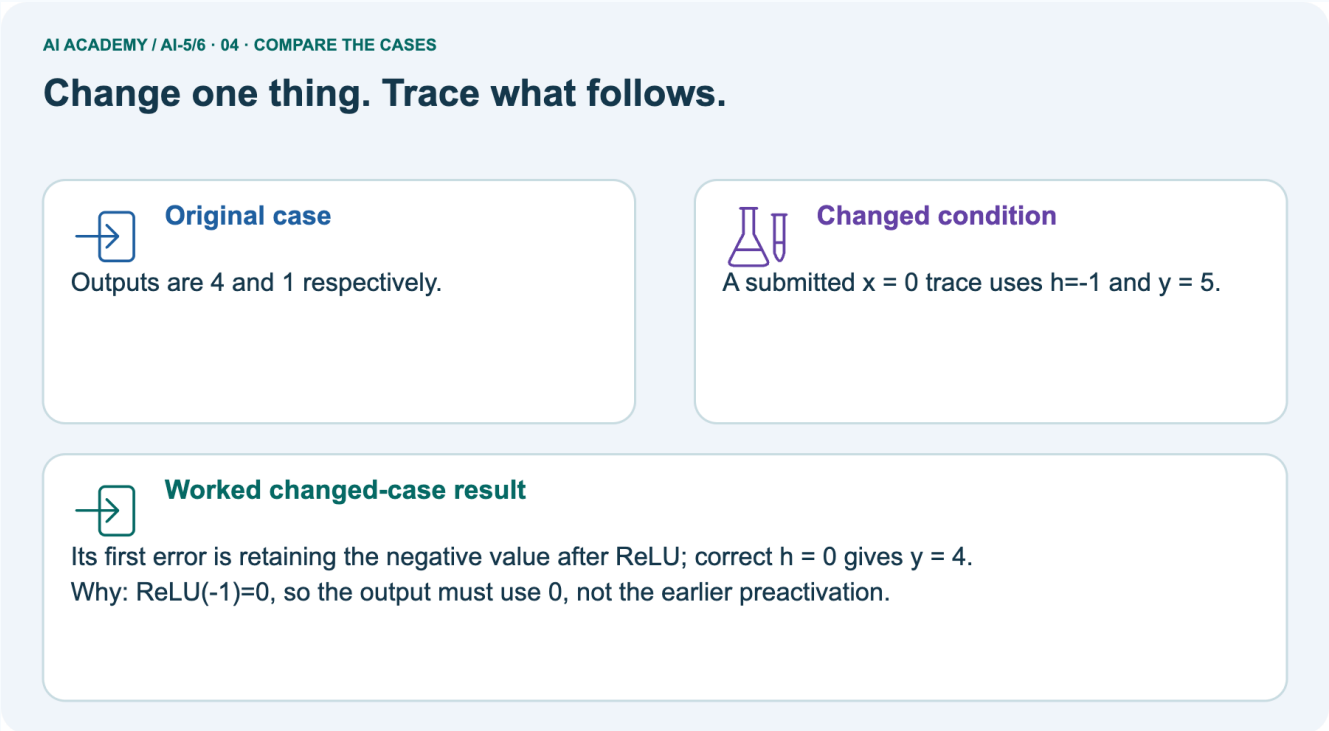
A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Predict the submitted scalar error

- Change one thing: A submitted x = 0 **trace** uses h=-1 and y = 5. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why the intermediate repair comes first

- A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

- Read the labelled picture: Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Separate main two-unit correct/submitted comparison

Main model x0 stage	Correct trace	Submitted missing-ReLU trace	Earliest difference?
Scores z	[−1, 2]	[−1, 2]	No
Hidden activations h	[0, 2]	[−1, 2]	Yes:negative first score was not clipped
Output y=h1−2h2+1	-3	-4	Later consequence; recompute after h repair

Correct a trace from the earliest error

TRACE STAGE AT X IS 0	EXPECTED VALUE	SUBMITTED VALUE
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Try the main x2boundary

Trace x = 2, recording both scores, both activations and y. What happens to the second unit at its zero-score boundary?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Forward passes by hand

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Choose tests that can expose the suspected bug

> A symmetric or all-positive case may hide an incorrect implementation.

Negative-score,boundary,bias and order-sensitive checks

A test where both scores are positive cannot reveal a missing ReLU. Include x = 0, where the first score is negative, and x = 3, where the second score is negative.

Include x = 0.5 and x = 2 to exercise each hidden zero-score boundary. Neither a few examples nor an attractive diagram proves every implementation case is correct.

A missing **output** bias shifts every **output** by -1. A swapped hidden order may pass a case where h1 equals h2; x = 1 has this limitation.

Keep several checks with different intermediate values, and retain them when repairing code.

Trace both hidden zero boundaries

- **Trace** x = 0.5 and x = 2. Which hidden score is zero in each case?

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise first-error diagnosis with less help

> Retain the submitted trace and the correction.

Current task and response

- **Trace** x = -1 through the lesson network. Record z, h and y.

> **Trace x = 2, recording both scores, both activations and y. What happens to the second unit at its zero-score boundary?**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “Forward passes by hand”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Forward passes by hand. Proposed explanation: “ReLU means the **output** cannot be negative”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured earliest-error reasoning

- Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent new network

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Locate a bias omission

Check your reasoning

Explain the picture

Run a complete first-error and regression clinic

> Every submitted trace,expected correction and discriminating test is supplied.

Current task and runnable forward trace

At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

```
def trace(x):
    z = [2 * x - 1, -x + 2]
    h = [max(0, value) for value in z]
    y = h[0] - 2 * h[1] + 1
    return z, h, y

for x in [0, 1, 3]:
    print(x, trace(x))
```

Four complete submitted traces and repair keys

Complete submitted main-network trace	Find first error; repair later values
x0:z[−1, 2], h[−1, 2], y−4	Activation first:h[0, 2], then y−3
x3:z[5, −1], h[5, 0], y5	Output first:missing bias+1; correct y6
x0:z[−1, −2], h[0, 0], y1	Score z2 first:−0+2=2; correct z[−1, 2], h[0, 2], y−3
x1:z[1, 1], h[1, 1], y0	This trace is correct but equal hidden values cannot expose a swapped-order implementation

Normal,boundary,missing and order-sensitive probes

Complete regression probe	Expected main trace / action
Normal x0 and3	x0:z[−1, 2], h[0, 2], y−3; x3:z[5, −1], h[5, 0], y6
Boundary x.5	z[0, 1.5], h[0, 1.5], y−2; first hidden score zero
Boundary x2	z[3, 0], h[3, 0], y4; second hidden score zero
Missing xNone	Outside numeric input contract; request/validate before trace
Swap hidden order at x0	Using[2, 0]instead of[0, 2]gives y3 instead of−3; this input reveals the bug
One final output matches	Still inspect all intermediates and retain varied regression cases; no general correctness or task-accuracy proof

Predict,audit,repair and retain regression cases

1. Audit all four supplied submitted traces before checking their repairs. Use the five independent expected rows to test the supplied trace function, then retain those expectations after a repair. Record the first incorrect stage and recompute its descendants; do not rewrite a final answer while keeping a wrong hidden value. Run or paper-trace each probe and label what you actually did.

   The code assumes numeric x and does not validate missing values. These examples verify fixed arithmetic, not learned usefulness or prediction accuracy.

2. On paper, **trace** the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit a different network’s submitted trace

> Use the new formulas and correct every value after the first error.

Complete new independent model and submitted trace

Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error

and correct all later values.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a different network independently

Attempt this without hints

Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error and correct all later values.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep expected and actual traces in the project record

> Preserve the failure and why the correction works.

Project artifact and evidence table

Add a labelled forward passes by hand **evidence** sheet to Image-pattern Network Investigation. Show how you can record scores and activations in **dependency** order.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a **trace** is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one first-error explanation and one revealing test

> A clean output alone does not tell the whole story.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Record scores and activations in **dependency** order

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Trace	A record of intermediate values in a computation
Preactivation	A weighted score before an activation function
Dependency	An earlier value needed to compute a later value
Regression check	A retained test rerun after a change to detect a new error

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- **Trace** x = -1 through the lesson network. Record z, h and y.
- **Trace** x = 0.5 and x = 2. Which hidden score is zero in each case?
- Explain why testing only x = 1 in the lesson model may miss a hidden-order swap.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent two-unit forward-audit model for the first individual attempt.

Current explanation and response

- A forward **trace** verifies what specified parameters calculate for one **input**. It does not show how those parameters were learned or whether the **output** is useful.

  Classification accuracy also needs a declared class rule, true labels and suitable independent examples.

- Next week separates the network architecture from its numerical parameters and counts the values that **training** could change.

- Explain why testing only x = 1 in the lesson model may miss a hidden-order swap.

A program matches y at x = 1 but its hidden activations are wrong. Is the implementation verified? Explain what evidence to retain after fixing it.

My limited claim and a useful next check

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Correct a trace from the earliest error

TRACE STAGE AT X IS 0	EXPECTED VALUE	SUBMITTED VALUE
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Audit a forward pass by writing every value at its correct stage.

By the end, I can	How I will show it
Record scores and activations in dependency order	Explain the deciding step, show a trace or test, and state one limit.
Locate the first incorrect value in a forward trace	Explain the deciding step, show a trace or test, and state one limit.
Separate a final output from evidence of model accuracy	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Audit a forward pass by writing every value at its correct stage.
- Which labelled step matches this detail? **Output**s are 4 and 1 respectively.
- 1. **Preactivation**
- 2. **Output**
- 3. Activation
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: **Output Output**s are 4 and 1 respectively. A forward pass applies a fixed network to an **input**; it does not update weights.

Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns.
For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Changed-case question and optional hint

- Changed-case practice: A submitted x = 0 **trace** uses h=-1 and y = 5. Which conclusion is supported by the supplied facts?
- 1. Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4. ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Repair and retain the record

Review Composing layers (ai-5/5). Goal: **Trace** how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a **trace** rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

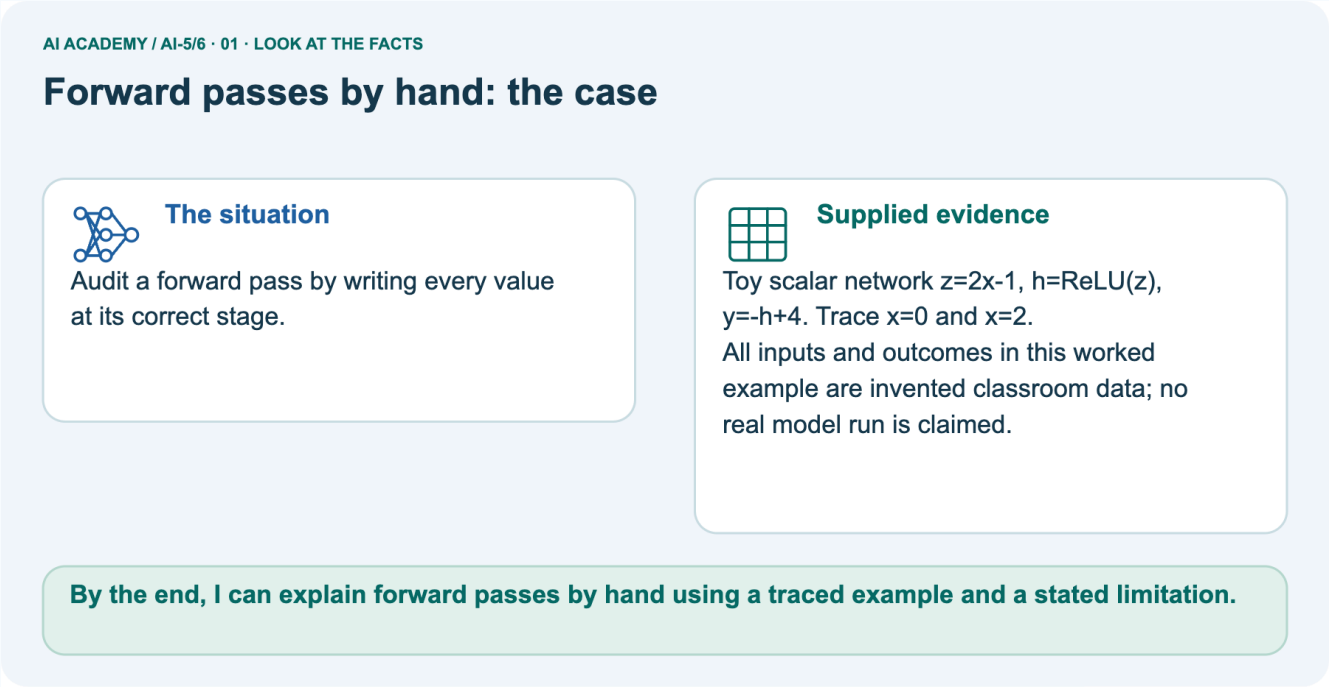
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Forward passes by hand: the case



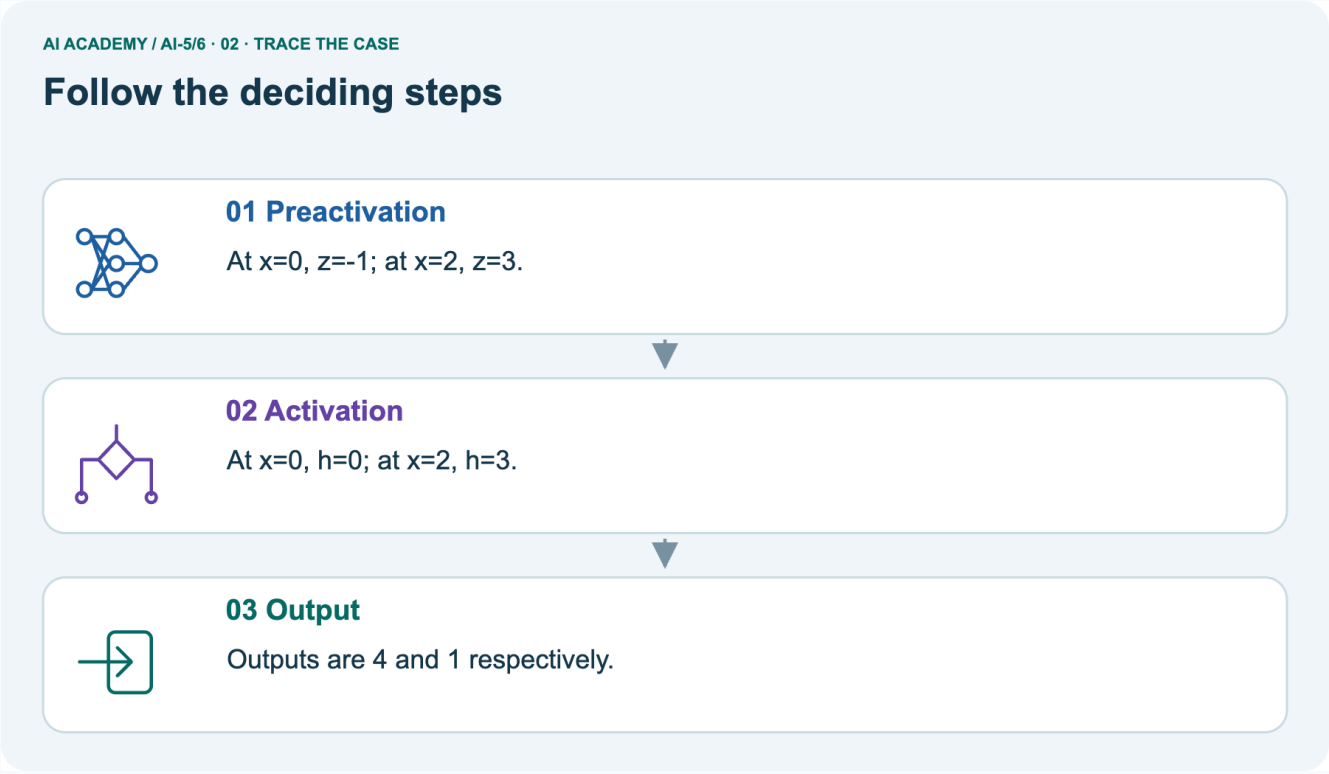
Follow the labelled picture with the matching explanation. Open the vector diagram

Audit a forward pass by writing every value at its correct stage. Supplied facts: Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Preactivation: At x=0, z=-1; at x=2, z=3. → Activation: At x=0, h=0; at x=2, h=3. → **Output**: **Output**s are 4 and 1 respectively.

Point to each numbered stage and explain which recorded **input** it uses.

03. Correct a trace from the earliest error

AI ACADEMY / AI-5/6 · 03 · INSPECT THE LESSON MODEL

Correct a trace from the earliest error

Trace stage at x is 0	Expected value	Submitted value
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

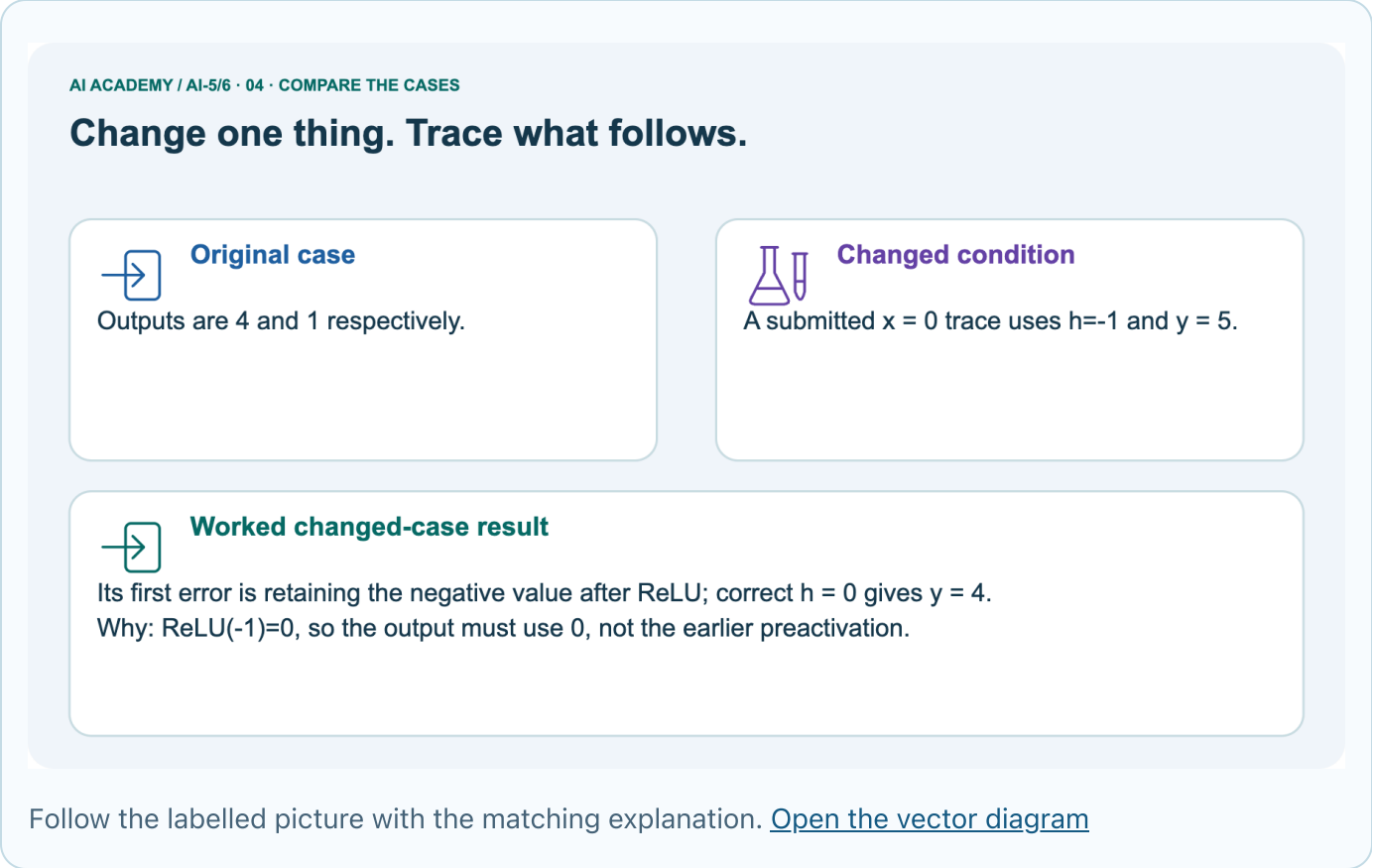
The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Follow the labelled picture with the matching explanation. Open the vector diagram

Correct a **trace** from the earliest error. Scores | [-1, 2] | [-1, 2]. ReLU activations | [0, 2] | [-1, 2]. **Output** | -3 | -4

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



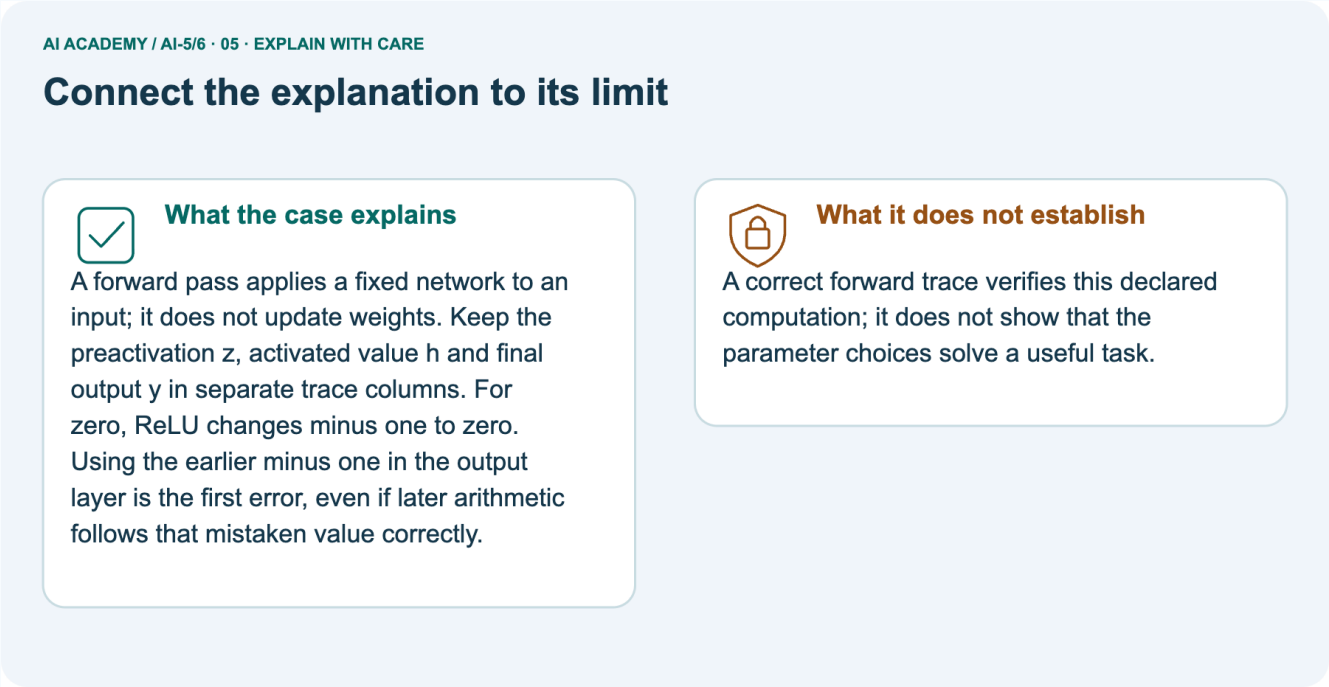
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Output**s are 4 and 1 respectively. Changed condition: A submitted x = 0 **trace** uses h=-1 and y = 5. Expected result: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

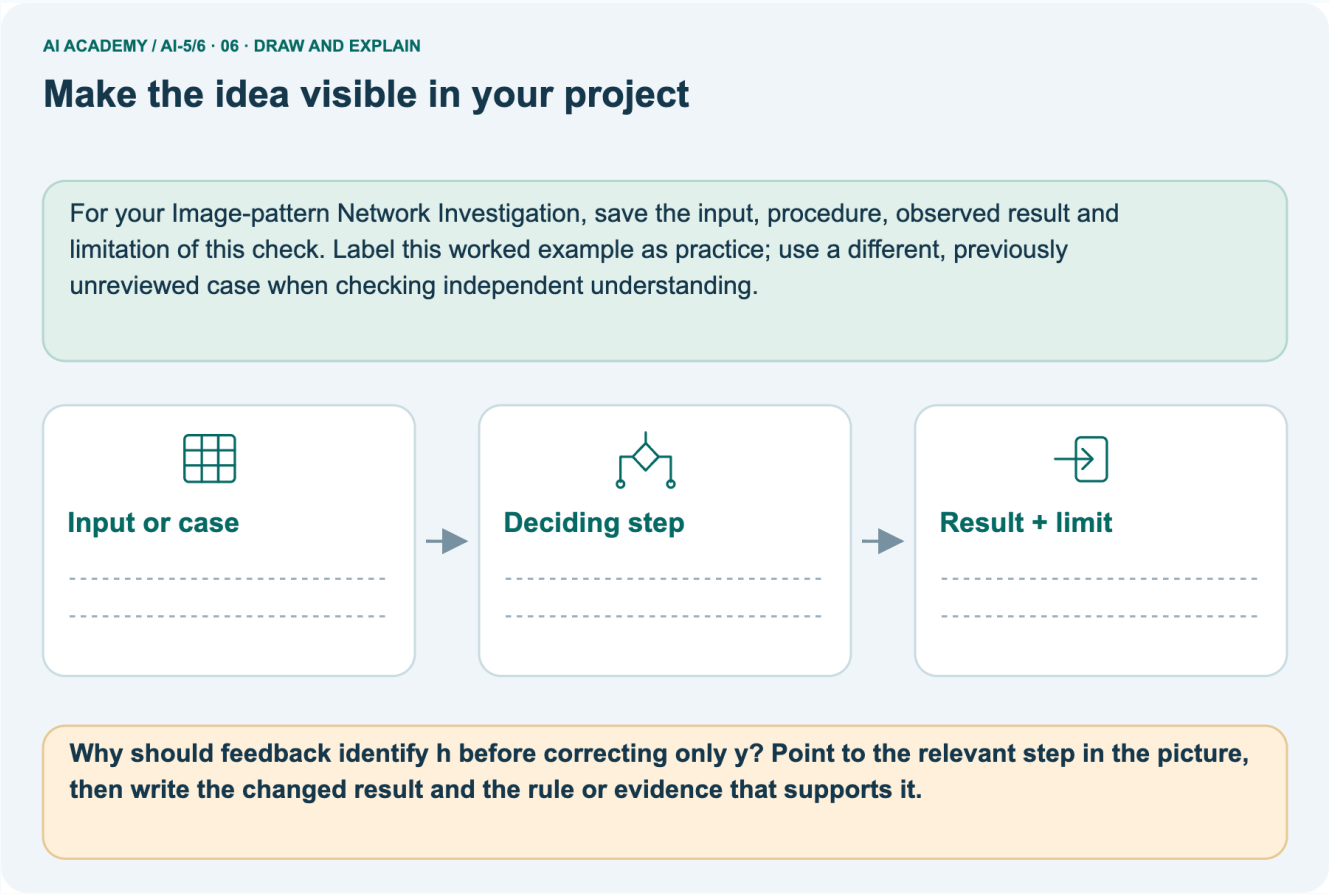
A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Limit: A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project **trace** and justify the deciding step using **evidence**.

Earlier wording, alternatives and picture transcripts

01. Forward passes by hand: the case

02. Follow the deciding steps

03. Correct a trace from the earliest error

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.